Multi-Agent Case Report

Case: demo_case_ecg00001hr_mriPatient001

Contributing agents: ecg, mri

Note: This case pairs a PTB-XL ECG record with an ACDC MRI patient. These datasets share no real patients - this is a synthetic pairing for multi-agent architecture validation, not a real joint clinical case.

Integrated Summary (Sep Agent)

The ECG shows a regular rhythm with an estimated mean heart rate of 63.9 bpm and a left ventricular ejection fraction (LVEF) of 23.7%. The MRI indicates low volumes in the left ventricle at both end-diastole and end-systole, as well as low volumes in the right ventricle at both end-diastole and end-systole. A short axis image shows a low value for the left ventricle.

Flags for review: low_lv_ejection_fraction_suspected

Traceability

Statement	Agent	Feature
The ECG shows a regular rhythm	ecg	rhythm_irregularity_flag
The ECG estimates a mean heart rate of 63.9 bpm	ecg	heart_rate_mean_bpm
The LVEF is estimated to be 23.7%	mri	lv_ejection_fraction_pct
Left ventricular volume at end-diastole is low	mri	lv_volume_ed_ml
Left ventricular volume at end-systole is low	mri	lv_volume_es_ml
Right ventricular volume at end-diastole is low	mri	rv_volume_ed_ml
Right ventricular volume at end-systole is low	mri	rv_volume_es_ml
Myocardial volume at end-diastole is low	mri	myocardium_volume_ed_ml
Myocardial volume at end-systole is low	mri	myocardium_volume_es_ml
Left ventricular ejection fraction is suspected to be low	mri	lv_ejection_fraction_pct

Note: traceability confirms each statement cites a real finding from a contributing agent - it does not verify that the summary's characterization of that finding (e.g. "low" vs "high") is clinically correct.

ECG Agent Findings

Record: 00001_hr
Patient: 56-year-old female, 63kg.

Flags: none

Feature	Value	Confidence	Evidence type	Rationale
duration_s	10.0	1.00	numeric_feature	Analyzed ECG trace duration is 10 seconds.
qrs_count	11	1.00	numeric_feature	Number of detected QRS complexes is 11.
heart_rate_mean_bpm	63.9	1.00	numeric_feature	Estimated mean heart rate is 63.9 bpm.
heart_rate_std_bpm	1.4	1.00	numeric_feature	Standard deviation of heart rate is 1.4 bpm.
rr_mean_ms	938.8	1.00	numeric_feature	Mean RR interval is 938.8 ms.
rr_std_ms	21.5	1.00	numeric_feature	Standard deviation of RR interval is 21.5 ms.
qrs_duration_mean_ms	59.3	1.00	numeric_feature	Mean QRS duration is 59.3 ms.
qrs_duration_std_ms	46.2	1.00	numeric_feature	Standard deviation of QRS duration is 46.2 ms.
rhythm_irregularity_flag		1.00	numeric_feature	Rhythm is regular.

MRI Agent Findings

Record: patient001
Patient: age/sex not available, 184cm/95kg.

Flags: low_lv_ejection_fraction_suspected

Feature	Value	Confidence	Evidence type	Rationale
lv_volume_ed_ml	295.5	1.00	numeric_feature	Left ventricular volume at end-diastole.
lv_volume_es_ml	225.6	1.00	numeric_feature	Left ventricular volume at end-systole.
rv_volume_ed_ml	139.7	1.00	numeric_feature	Right ventricular volume at end-diastole.
rv_volume_es_ml	59.5	1.00	numeric_feature	Right ventricular volume at end-systole.
myocardium_volume_ed_ml	164.3	1.00	numeric_feature	Myocardial volume at end-diastole.
myocardium_volume_es_ml	195.1	1.00	numeric_feature	Myocardial volume at end-systole.
lv_ejection_fraction_pct	23.7	1.00	numeric_feature	Left ventricular ejection fraction.
rv_ejection_fraction_pct	57.4	1.00	numeric_feature	Right ventricular ejection fraction.
short_axis_mid_ed		1.00	image_region	Short axis image at end-diastole.

Figures

C:\Users\neifa\Documents\ENETCOM\3IDSD\stage_25-26\ECG\results\records\patient001\mri_pipeline\patient001_ed_es_roi.png

